

Does Cypress Score Over Selenium for Web App Testing?

A decade ago, the Web wasn't what it is today. The changing Web demands a change in testing tool! Testing is a critical process in application development and a project's success or failure is entirely dependent on it. This article takes the reader through Cypress – a tool for Web app testing.



Website testing is entirely different from orthodox software testing. There are some things that make Web testing more challenging for quality analysis (QA), such as dealing with XHR calls and Web services, security of data, dynamic behaviour of the application due to modern development frameworks, and so on. Selenium has been a chief performer in E2E (exchange-to-exchange) Web application testing for years now, but the current Web is different. Though Selenium is a very established testing tool, it's hard to set it up, as other libraries need to be installed with it. Debugging tests are also not carried out upfront, which may slow down the testers. Cypress overcomes these limitations of Selenium.

Getting to know Cypress

Cypress is a fairly new player in the field of automation and Web app testing. Just like the Selenium framework, Cypress is an open source tool and is also hosted on GitHub. It is a JavaScript based end-to-end testing framework that does not use Selenium in any way. It's built on top of Mocha, which is again a feature-rich JavaScript test framework running on and within the browser, making asynchronous testing simple and fun. Cypress also uses a BDD/TDD assertion library and a browser that can be paired with any JavaScript testing framework.

Prerequisites: To work with Cypress we should have knowledge of JavaScript and have Node.js installed in our system.

Why Cypress?

The developer of Cypress, Brian Mann, did a survey to find out the challenges being faced in testing. He tried to tackle most of the shortcomings of earlier testing tools by developing Cypress. Although Cypress has many advantages, I would like to highlight two that I found fascinating.

- **Automatic waiting:** Cypress automatically waits for the DOM to load, elements to become noticeable, the animation to be completed, the XHR and AJAX calls to be finished, and much more. Hence, there's no need to define implicit and explicit waits.
- **Real-time reloads:** Cypress is intelligent enough to recognise that after saving the test file (*xyz_spec.js file*), it will be run again; so it automatically triggers the run next to the browser as soon as the file is saved. Hence, there is no necessity to manually trigger the run.

Who uses Cypress?

Cypress is typically used by front-end developers or QA engineers building Web applications using modern JavaScript frameworks.

It enables us to write various types of tests:

- End-to-end tests
- Integration tests
- Unit tests

Cypress can test everything that runs on a browser.